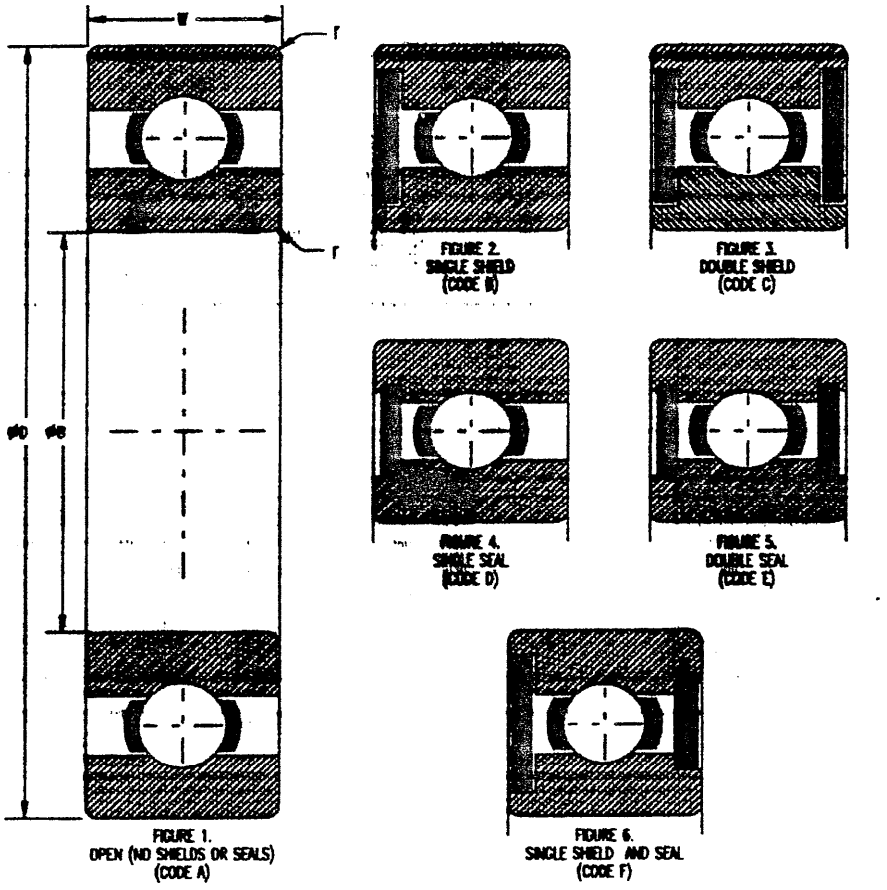
METRIC
FF-B-171/4
30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, SINGLE ROW, RADIAL, NON-FILLING SLOT, ANSI/AFBMA DIMENSION SERIES 02.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.



BEARING CONFIGURATIONS

AMSC/NA FSC 3110

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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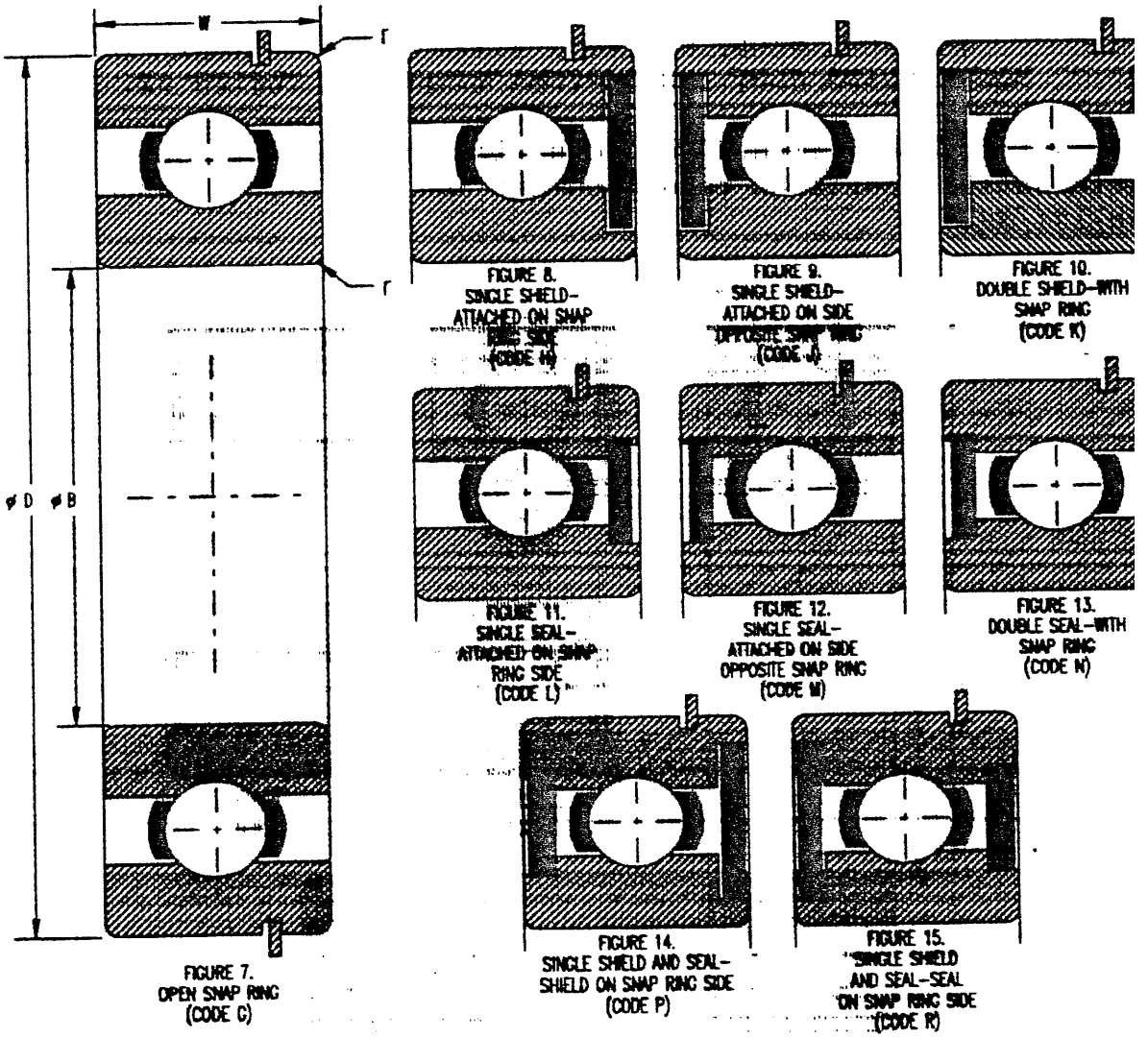


FIGURE 7.
OPEN SNAP RING
(CODE C)

FIGURE 8.
SINGLE SHIELD-
ATTACHED ON SNAP
RING SIDE
(CODE H)

FIGURE 9.
SINGLE SHIELD-
ATTACHED ON SIDE
OPPOSITE SNAP RING
(CODE J)

FIGURE 10.
DOUBLE SHIELD-WITH
SNAP RING
(CODE K)

FIGURE 11.
SINGLE SEAL-
ATTACHED ON SNAP
RING SIDE
(CODE L)

FIGURE 12.
SINGLE SEAL-
ATTACHED ON SIDE
OPPOSITE SNAP RING
(CODE M)

FIGURE 13.
DOUBLE SEAL-WITH
SNAP RING
(CODE N)

FIGURE 14.
SINGLE SHIELD AND SEAL-
SHIELD ON SNAP RING SIDE
(CODE P)

FIGURE 15.
SINGLE SHIELD
AND SEAL-SEAL
ON SNAP RING SIDE
(CODE R)

BEARING CONFIGURATIONS - (Continued)

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REQUIREMENTS:

1. **Boundary dimensions.** The boundary dimensions shall be specified in the part number by the size codes in Table I.

2. Cages. Cages shall be manufactured groin any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Shields, seals, **and snap** rings. Shields and seals shall be fabricated from any material unless otherwise specified and shall be removable or nonremovable type. Seals shall be impervious contact type. Snap rings shall be fabricated from ferrous material unless otherwise specified and shall have sufficient spring temper to set firmly in the ring grooves and shall be removable without the metal taking a permanent set. The snap ring and snap ring groove dimensions and their relative locations shall be in accordance with ANSI/AFBMA Standard 20. The number and positioning of shields, seals, and snap rings required shall be specified in the part number by the codes in table III.

4. **Radial Internal** Clearance. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

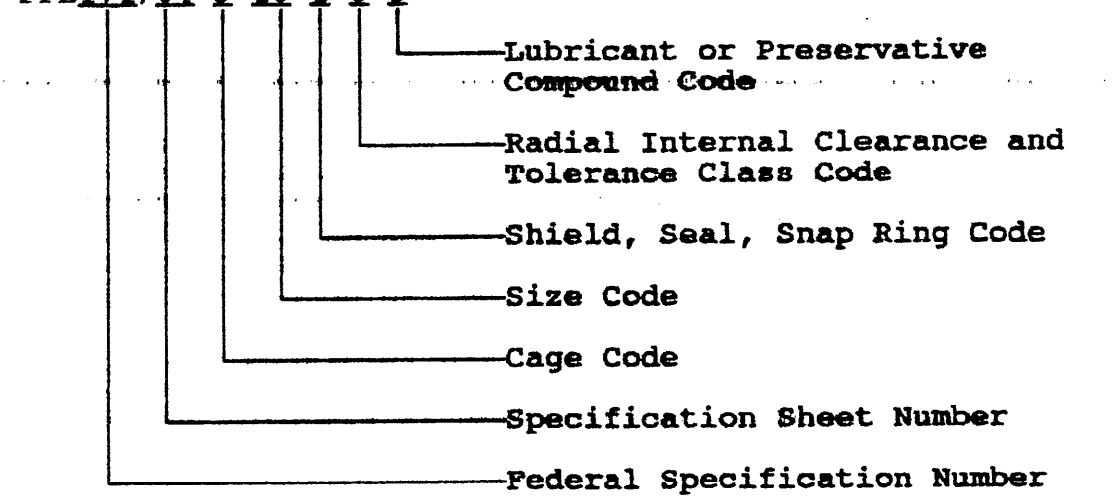
5. **Tolerance Class.** The tolerance limits for tolerance class ABEC-1 arid ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

6. **Lubricant or Preservative** Compound. Preservative compound shall be applied to all open bearings (no shields or seals) and to single closure bearings (single shield or single seal) that have no additional lubricant requirement. Thus, for these bearings the bearing part number shall include the “C” suffix. For double closure or single closure bearings that require lubricant the bearing part number shall include the “A” or “B” suffix. When grease is required, the bearing void (the airspace between the inner and outer ring of the assembled bearing) shall be 25 to 50 percent filled. The lubricant or preservative compound shall be specified in the part number by the codes in table V.

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7. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, shields, seals, and snap rings, radial internal clearance and tolerance class, and lubricant or preservative compound.

EXAMPLE: FFB171/04 J 10 G 3 C



FFB171/04J10G3C indicates - Bearing, cage-pressed steel or molded plastic; bore 50mm; outside diameter 90mm; width 20mm; open (no shields or seals) with snap ring; radial internal clearance symbol 3 [greater than normal); tolerance class ABEC-1; preservative compound in accordance with MIL-P-197.

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TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic load rating (pounds)
00	10	30	9	0.6	15	805
01	12	32	10	0.6	17	1175
02	15	35	11	0.6	20	1320
03	17	40	12	0.6	22	1655
04	20	47	14	1.0	26	2220
05	25	52	15	1.0	31	2435
06	30	62	16	1.0	36	3380
07	35	72	17	1.1	42	4460
08	40	80	18	1.1	47	5060
09	45	85	19	1.1	52	5680
10	50	90	20	1.1	57	6090
11	55	100	21	1.5	63.5	7530
12	60	110	22	1.5	68.5	9110
13	65	120	23	1.5	73.5	9935
14	70	125	24	1.5	78.5	10800
15	75	130	25	1.5	83.5	11480
16	80	140	26	2.0	90	12610
17	85	150	28	2.0	95	14445
18	90	160	30	2.0	100	16665
19	95	170	32	2.1	107	18880
20	100	180	34	2.1	112	21215
21	105	190	36	2.1	117	23100
22	110	200	38	2.1	122	25025
24	120	215	40	2.1	132	25145
26	130	230	40	3.0	144	26995
28	140	250	42	3.0	154	28805
30	150	270	45	3.0	164	30470
32	160	290	48	3.0	174	32005
34	170	310	52	4.0	188	36725
36	180	320	52	4.0	198	39365
38	190	340	55	4.0	208	44135

1/ Size codes in Table I are industry bore codes.

2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.

2/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

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TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)

TABLE III. Codes for Shields, Seals, and Snap Rings.

Code	Shield, seal, and snap ring configuration
A	open (no shields or seals) - fig.1
B	single shield - fig.2
C	double shield - fig.3
D	single seal - fig.4
E	double seal - fig.5
F	single shield and seal - fig.6
G	open with snap ring - fig.7
H	single shield (attached on snap ring side) - fig.8
J	single shield (attached on side opposite snap ring)-fig.9
K	double shield with snap ring - fig.10
L	single seal (attached on snap ring side) - fig.11
M	single seal (attached on side opposite snap ring)-fig.12
N	double seal with snap ring - fig.13
P	single shield and seal (shield on snap ring side)-fig.14
R	single shield and seal (seal on snap ring side) - fig.15

TABLE IV. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (less than normal)	ABEC-1
2	Symbol 0 (normal)	
3	Symbol 3 (greater than normal)	
4	Symbol 4 (greater than symbol 3)	
5	Symbol 2 (less than normal)	ABEC-5
6	Symbol 0 (normal)	
7	Symbol 3 (greater than normal)	
8	Symbol 4 (greater than symbol 3)	

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TABLE V. Codes for Lubricant and Preservative Compound.

Code	Lubricant or Preservative Compound
A	Grease in accordance with MIL-G-81322
B	Grease in accordance with DOD-G-24508
C	Preservative compound in accordance with MIL-P-197

MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army - AT
Air Force - 11

Review Activities:
Army - AR, EA, MI
Navy - OS

User Activity:
Army - GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

PREPARING ACTIVITY:
DLA-IS

(Project 3110-0848-04)

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